REPUBLIQUE TUNISIENNE
Ministère de l'Enseignement Supérieur
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Concours Nationaux d'Entrée aux Cycles de
Formation d'Ingénieurs
Session 2020

الجمهورية التونسية
وزارة التعليم العالي
والبحث العلمي
المناظرات الوطنية للدخول
إلى مراحل تكوين المهندسين
دورة 2020

Concours Toutes Options
Epreuve d'Anglais

Date : 22 Juillet 2020 **Heure : 11 H** **Durée : 2 H** **Nb pages : 08**

Barème : Part I : 30, Part II : 30, Part III : 20

IMPORTANT :

1- ***L'épreuve d'anglais comporte deux séries de feuilles :***

- Les énoncés s'étalant sur 4 pages que les candidats sont appelés à garder

- Les feuilles réservées aux réponses (Answer sheet) s'étalant sur 4 pages, lesquelles

Doivent être rendues à la fin de l'épreuve aux professeurs surveillants

2- ***Il sera tenu compte de la présentation, (l'écriture au crayon n'étant pas permise)***

Reading passage:

1- There is a technological juggernaut heading our way. It's called the Internet of Things (IoT). For the tech industry, it's the Next Big Thing, alongside big data, though in fact that pair are often just **two sides of the same coin**. The basic idea is that since computing devices are getting smaller and cheaper, and wireless network technology is becoming ubiquitous, it will soon be feasible to have trillions of tiny, networked computers embedded in everything. They can sense changes, turning things on and off, making decisions about whether to open a door or close a valve or order fresh supplies of milk, you name it, the computers communicating with one another and shipping data to server farms all over the place.

2- As ever with digital technology, there's an underlying rationality to lots of this. The IoT could make our lives easier and our societies more efficient. If parking bays could signal to nearby cars that they are empty, then the nightmarish task of finding a parking place in crowded cities would be eased. If every river in the UK could tweet its level every few minutes, then we could have advance warning of downstream floods in time to alert those living in their paths. And so on.

3- But that kind of networking infrastructure takes time to build, so the IoT boys have set their sights closer to home, which is why we are beginning to hear a lot about "smart" homes. On further examination, this turns out mostly to mean houses stuffed with networked kit.

4- "Because every home should be a smart home this Christmas," burbles an ad from Samsung, "we're treating you to some fantastic deals. Getting started is easy with SmartThings. Simply set up your SmartThings Hub and free app, add in your favourite products, and take control from another room -or another country. It's simple to set up, too, meaning you can start making your home smarter in no time". The smart things in question include sensors of various kinds, remotely controlled power outlets, a wireless hub and -of course- a smartphone app that enables you to turn on your lights (or see who's in your sitting room) when you're on a beach in Thailand.

5- And it's not just Samsung. Other big companies are getting in on the act. Google, for example, has bought Nest, an outfit that makes a "learning thermostat" that uses sensors and algorithms to learn about your energy usage and save you money. A study by Accenture found that 13% of homes already have one or more IoT devices and predicts that 69% will have an IoT device by 2019. Home-owners who place too much trust in such devices could easily wind up on the PleaseRobMe website. As with all such predictions, this one should be taken with a grain of salt. But suppose, for a moment, that it's accurate and that within a decade a large proportion of homes have become smart. What would be the implications?

6- To qualify as a smart home, a house would have to have multiple devices, most of which would be linked wirelessly to a hub that in turn communicates with the outside world. The devices will monitor the activities and rhythms of the household so that they can optimise energy consumption and so on. That means data flowing from devices to other devices via the hub and perhaps also to external servers, suppliers etc. So a smart home monitors- or, you could say, spies on – its owner(s). And that could have serious implications for privacy.

7- Further, a smart home is, by definition, a networked one. And that, in turn, has consequences. One is increased complexity – and corresponding user bafflement if anything goes wrong. Another is insecurity, since nothing connected to the internet can be regarded as totally secure. And this is deeply ironic, since we know that one of the reasons why people want smart homes – with all the associated monitoring enabled by the technology – is to increase their security.

8- So-called smart meters provide an intrusive case study of this paradox in action. At the moment, most of our electricity and gas meters are dumb, analogue devices: they record your consumption and someone comes round periodically to take a reading. This is inefficient and wasteful and so there's a strong push under way to fit homes with smart meters. These are purely digital devices that are networked so that the utility company can bill you without having to send a man to your door.

9- The smart meter has advantages for consumers. Since the price of electricity varies on an hourly basis, the meter enables them to decide when they wish to consume it. They can, for example, run their washing machines and dishwashers in the middle of the night, when demand – and therefore the price per unit – is low. And for the utility company, the meter has significant benefits, such as being able remotely to disconnect consumers' supply if they don't pay the bill, instead of going to all the legal and logistical palaver currently involved in cutting off a subscriber.

10- But there is an obvious fly in this soothing ointment. For if the utility company can remotely disconnect you, then, given that there is no such thing as a completely secure networked device, so could a hacker in Novosibirsk. In fact, he could conceivably disconnect every house in East Anglia. It's the old, old story: **technology giveth and technology taketh away.**

John Naughton
The Guardian Sat 2 Dec 2017

Part ONE READING COMPREHENSION (30 marks)

I- Choose the best option to be the title of the article: (1mark)

1- Should we be worried if our homes are smarter than we are?

2- Should we be delighted if a technological juggernaut is heading our way?

3- Should smart homes be our future?

II- With reference to the passage, complete the chart by focusing on the benefits the IoT could provide in the following fields to facilitate citizen's life: (3marks)

The field	The benefits
- Traffic	.....
- Civil Protection	.....
- Energy	.....

III- Answer the following questions: (7marks)

1- Which technological components are mainly needed to make homes smart? (4marks)

2- What are the drawbacks people may face in case they opt for a smart home? (3marks)

IV- Complete with words or phrases with reference to the text: (4marks)

1- Both … (a)… as well as the utility company can benefit from the smart meters' installation in homes. This enables the latter not only to …(b)… but also …(c)...

2- Homes that are fit with networked computers become smart, yet there's no guarantee that these houses are …(a)…. and that they wouldn't be broken into.

V- Say whether the following statements are true or false and justify from the text: (4marks)

1- The digital technology is so flourishing that it is expected to have almost two thirds of users sharing IoT devices in their homes.

2- The price of electricity is uniform.

VI- Select the right option: (3marks)

1- Could easily wind up on the ''please RobMe website'' (§5- line 5) means:

a- Could be accused of robbery.

b- Could find yourself in an unexpected and unpleasant situation.

c- Could become hooked on a website.

2- Should be taken ''with a grain of salt'' (§5- line 6) means:

a- Accept with reservation.

b- Accept with admiration.

c- Accept blindly.

3- There's an obvious fly in this soothing ointment (§10- line 1) means:

a- There's a minor irritation that spoils enjoyment.

b- There's a flaw that can be corrected quickly.

c- There's a grave implication that is irreversible.

VII- Explain the following using your own words: (2marks)

1- two sides of the same coin (§1) 2- technology giveth and technology taketh away (§10)

VIII- What do the underlined words in the text refer to? (3marks)

1- that pair (§1) 2- the act (§5) 3- so (§10)

IX- Find words that have approximately the same meaning as the following: (3marks)

1- fundamental (§2) 2- company (§5) 3- precise (§5)

Part TWO LANGUAGE (30 marks)

I- Put the bracketed words in the right tense or form (10marks)

Wildfires are a feature of life in Australia, which is not surprising when you consider that it is the **[1] (dry)** inhabited continent in the world. Major fires often **[2] (occur)** at this time of year, but they have not been as **[3] (destroy)** as we **[4] (see)** right now. There **[5] (be)** many massive wildfire outbreaks since records **[6] (begin)** in 1851, and more than 800 people **[7] (know)** to have died. Throughout the 20th century these conflagrations **[8] (grow)** much worse. During this latest crisis, police in South Wales and Queensland **[9] (charge)** a number of people, not more than 100, with **[10] (deliberate)** starting fires.

II- Complete with the appropriate word: (The first two letters the word begins with and a synonym of the word are provided to help you) (10 marks)

Uber's service is built around their smartphone app used by both drivers and **[1] cu… (=clients)**, which gives them an **[2] op… (=chance)** to collect GPS, gyroscope and accelerometer data during Uber **[3] jo… (=trips)**. Data is **[4] co… (=continuously)** collected during trips and sent to Uber's servers for processing and long-term storage. Uber sends drivers comparative reports on their driving habits by analysing the trip data for rapid acceleration, harsh braking, speeding, or **[5] ri… (=dangerous)** cornering, but also stores data to find long-term driver- or location- **[6] pa… (=specific)** trends. The **[7] ob… (=goal)** of using telematics for Uber is to **[8] be…(=improve)** driver safety and **[9] re… (=decrease)** car crashes. At the same time, they can **[10] ga… (=collect)** important and useful information on roads' conditions and driver reactions.

III- Express differently starting as given on the answer sheet (keep the same meaning): (10marks)

1- I have never witnessed such a devastating hurricane.

2- It isn't necessary for you to remind me all the time to watch my language.

3- Why didn't you teach them how to use this machine?

4- They could hardly wait to move into their new house.

5- I'm sorry, I'd prefer not to go sailing now.

6- The president will not be re-elected if he doesn't reduce the deficit.

7- They claimed the road had been closed for repair.

8- "What did they steal?", asked the police officer.

9- The government didn't respond quickly to the outbreak that's why the virus affected so many.

10- Despite the U.N. Environment continuous support to African governments, they are unable to tackle the issue of electronic waste.

Part THREE WRITING (20 marks)

I- Re-arrange to obtain a coherent sentence (3marks)

a/ and killed more than 350.000 worldwide,

b/ which has infected at least 5 million people

c/ strain of the coronavirus disease,

d/ where the outbreak started.

e/ bats are known carriers of the latest

f/ mostly in China

II- Find the FOUR mistakes in the following sentence and correct them on the answer sheet: (2marks)

Airports cross the world are deploying screened measures to isolate passengers with symptoms related by the disease to contain it's spread.

III- Essay (15marks) (DO NOT EXCEED 15 LINES)

With the lockdown (confinement) taking place in many countries of the world, various positive impacts have been experienced. To what extent do you agree or disagree with this opinion? Give reasons for your answer and include any relevant examples from your own knowledge or experience.

Concours 2020 **Epreuve de : ANGLAIS** **SUGGESTED ANSWERS**

PART ONE Reading Comprehension (30marks)

I- Choose the best option to be the best title of the article: (1mark)

(1) Should we be worried if our homes are smarter than we are?

II- With reference to the passage, complete the chart by focusing on the benefits the IoT could provide in the following fields to facilitate citizen's life: (3marks)

The field	The benefits
- Traffic	***Signalling to nearby cars that parking bays are empty***
- Civil Protection	***Having advance warning of downstream floods***
- Energy	***Optimising/monitoring energy consumption*** Learning about your energy usage

III- Answer the following questions: (7marks)

1- Which technological components are mainly needed to make homes smart? (4marks)

(a) Sensors

(b) Remotely-controlled power outlets

(c) A wireless hub

(d) A smart app

2- What are the drawbacks people may face in case they opt for a smart home? (3marks)

(a) Privacy is at risk/ spying on the owners
(b) Complexity/ user bafflement if anything goes wrong
(c) Insecurity (PleaseRobMe website)

IV- Complete with words or phrases with reference to the text: (6marks)

1- Both the (a) ***consumer/subscriber/ owner*** (customer/house owner) as well as the utility company can benefit from the smart meters' installation in homes. This enables the latter not only to (b) ***bill the consumer without having to send a man to your door*** but also (c) ***remotely disconnect consumers' supply.***

2- Homes that are fit with networked computers become smart, yet there's no guarantee that these houses are (a) (***completely/totally/secure)*** (safe) and that they wouldn't be broken into.

V- Say whether the following statements are true or false and justify from the text: (4marks)

1- The digital technology is so flourishing that it is expected to have almost two thirds of users sharing IoT devices in their homes.

(true) A study by Accenture … predicts that 69% will have an IoT device by 2019. (§5)

2- The price of electricity is uniform.

(false) Since the price of electricity varies on an hourly basis/ when demand – and therefore the price per unit – is low. (§9)

VI- Select the right option: (3marks)

1- Could easily wind up on the ''please RobMe website'' (§5- line 5) means:

b- Could find yourself in an unexpected and unpleasant situation.

2- Should be taken ''with a grain of salt'' (§5- line 6) means:

a- Accept with reservation.

3- There's an obvious fly in this soothing ointment (§10- line 1) means:

a- There's a minor irritation that spoils enjoyment.

VII- Explain the following using your own words (2marks)

1- two sides of the same coin (§1): ***the same thing presented in two different ways***

2- technology giveth and technology taketh away (§10): ***as it has advantages, technology also has drawbacks***

VIII- What do the underlined words in the text refer to? (3marks)

1- that pair (§1) : ***the Iot (the Next BigThing), big data***

2- the act (§5): ***every home should be a smart home***

3- so (§10): ***remotely disconnect you*** / the ability of disconnecting you

IX- Find words that have approximately the same meaning as the following: (3marks)

1- fundamental (§2): ***underlying***

2- company (§5): ***outfit***

3- precise (§5): ***accurate***

PART TWO LANGUAGE (30marks)

I- Put the bracketed words in the right tense or form (10marks)

1	**driest**	**6**	**began**
2	**occur**	**7**	**are known**
3	**destructive**	**8**	**grew**
4	**are seeing**	**9**	**have charged**
5	**have been**	**10**	**deliberately**

II- Fill in the blanks with the appropriate words: (10marks)

customer (1/2)

1	**customers**	**6**	**particular**
2	**opportunity**	**7**	**objective**
3	**journeys**	**8**	**better**
4	**constantly/continually** consistently	**9**	**reduce/ restrain**
5	**risky**	**10**	**gather**

III- Express differently starting as given (keep the same meaning): (10marks)

1- This is ***the most devastating hurricane I have ever witnessed.***

2- I (***needn't be / don't have to be/ don't need to be)*** reminded all the time to watch my language.

3- You should ***have taught them how to use this machine.***

4- They were looking forward ***to moving into their new house.***

5- I'm sorry, I'd rather ***not go sailing now.***

6- Unless ***he reduces the deficit, the president won't be re-elected.***

7- The road ***was claimed to have been closed for repair.***

8- The police officer asked ***what they had stolen.***

9- Had ***the government responded quickly to the outbreak, the virus wouldn't have affected so many.***

10- Although ***the U.N. Environment has continuously supported (continuously supports / is continuously supporting) the African governments, they are unable to tackle the issue of electronic waste.***

PART THREE WRITING (20 marks)

I- Re-arrange to obtain a coherent sentence: (write the corresponding letters) (3marks)

1→ *e* 2→ *c* 3→ *b* 4→ *a* 5→ *f* 6→ *d*

II- Find the FOUR mistakes and correct them: (2marks)

Mistake	Correction
cross	*across*
screened	*screening*
by	*to*
it's	*its*

III- ESSAY (15marks)

.....

Suggested marking scheme: 5 pts: ideas / 2 pts: grammar / 2pts: vocabulary /

2pts: spelling / 2 pts: cohesion & coherence/ 2pts: punctuation

.....

Suggested ideas in case of agreement with the statement :

1-Positive environmental aspects : *many species enjoying the extra space such as sea turtles, reptiles, dolphins ... kangaroos and goats have more freedom to roam * NASA and the European Space Agency first saw the effect via satellite data that shows China's dramatic drop in nitrogen dioxide emissions/ reduction in air pollution : the Himalayas are visible from parts of India... life everywhere just seems to be breathing easier.
2-Crises breed so much innovation: *innovators in different fields are jumping in to help * manufacturers shifted their production to hand sanitizers, face masks, life-saving machines..

* Engineers' newly-invented detectors are installed in all hospitals' entries

3-Positive social aspects : * people have become physically distant but socially / emotionally more connected (parents spending more time with their children). * Volunteeers have worked hard to reduce social isolation and loneliness by delivering emergency food parcels or medicines for people in need especially those who may be older and more at risk from corona virus . * Others have used their balconies in unconventional ways ; they stepped out to their balconies for fresh air , chats with neighbours and to cheer for healthcare workers * more interest in hygiene as a way of survival: no more shaking hands , covering mouth when coughing , wearing face masks in public places and sanitizing hands ...

4-Positive educational aspects Physical distancing has brought about new innovative volunteer opportunities and has motivated helpers to find plenty of creative ways to serve others remotely .Many teachers have given multiple courses to students online , encouraging them to keep learning in this crucial period .

5, Teleworking

Suggested ideas in case of disagreement with the statement :

Personally I disagree with the opinion / I find the opinion illogical ....for the reasons to come :
The economy has taken a big hit. *Small businesses which thrive on daily incomes have been badly shaken . Big businesses have not been spared either. *Tourism and travel have been rattled. Many airlines have already declared their situation as dire. Some have already filed for bankruptcy. Thousands of workers in different fields have been made **jobless**.
To contain the spread of the coronavirus, many countries have shut workplaces, schools and places of entertainment, confining millions to their home. This has affected **psychologically** many people, giving rise to the "cabin fever syndrome":claustrophobia... When stuck indoors, some people become what has been called "stir crazy" : restlessness, irritability, impatience, feelings of lethargy, difficulties concentrating, low motivation, food cravings (gaining weight) and sleep disorders (difficulties to fall asleep or sleeping too much).
Socially, the lockdown situation has given abusers free will to carry on with their misdeeds . Increased numbers of children and women have been exposed to domestic violence . In Tunisia the number of women suffering from domestic abuse has had a nine fold increase during the lockdown leading to divorce .The anxiety and the financial pressure are fuelling the stress factors in many families. "For some children and mothers ,home is not a place of safety."